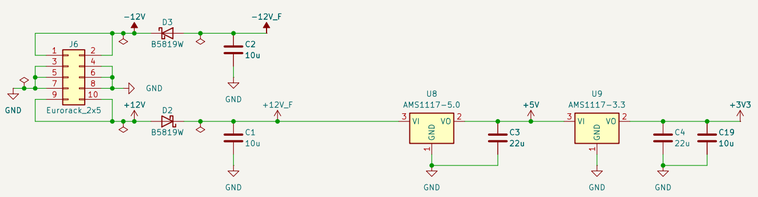
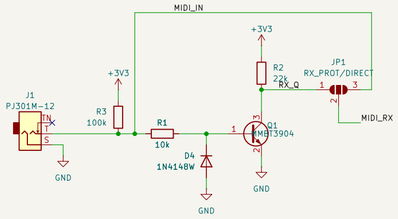
BIG MODULAR 32BIT — S1 schematic (Bread Modular 32bit–5v for Eurorack, 6HP, 4 layers). Circuit intent: docs/research.md; placement: docs/geometry.json

POWER: +12V -> D2 -> +12V_F (op-amp V+, U8 AMS1117-5.0 -> +5V -> U9 AMS1117-3.3 -> +3V3); -12V -> D3 -> -12V_F (op-amp V-)



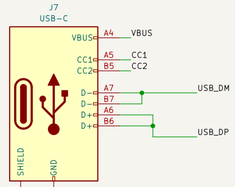
PWR_FLAG: GND/+12V/-12V mark the Eurorack entry at J6; +12V_F/-12V_F are fed only through D2/D3; +3V3_ESP32 (via FB1) and +3V3_8388 (via R5) are fed through passive parts.

MIDI IN (TS): modules/16bit NPN front end at 3.3 V, no opto -> RXD0 (GPIO44, UART0)

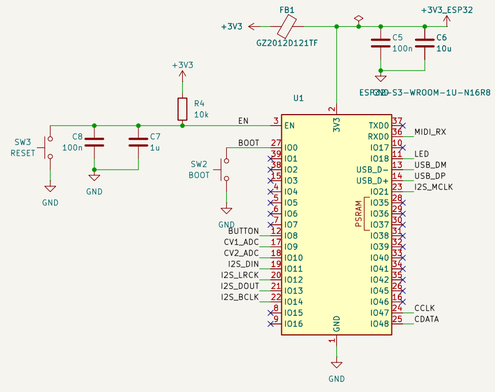


JP1 RX SOURCE (back, shipped OPEN, bridge one pair): 1–2 = PROTECTED (Q1 drives RXD0, firmware inverts the UART RX line); 2–3 = DIRECT (unprotected, stock BM firmware). Never both.

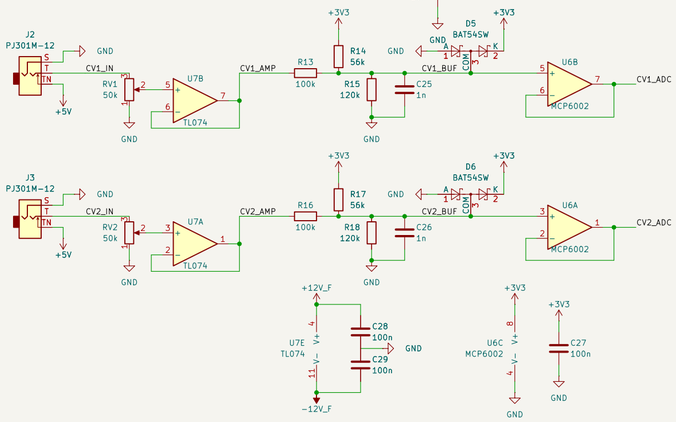
USB-C (data only, as Bread Modular): VBUS not connected, D+/D- straight to the ESP32-S3 USB PHY (flash over USB).



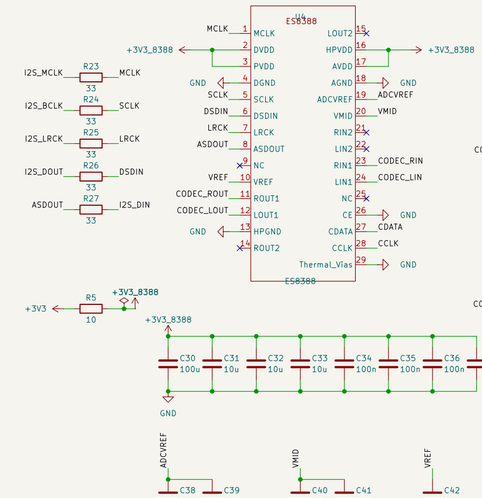
ESP32-S3-WROOM-1U-N16R8 (Bread Modular 32bit–5v GPIO map).
RESET = EN low; BOOT = IO0 low at reset (download mode). IO35–37 = octal PSRAM, keep open.



CV IN 1/2: tip normalled to +5V -> RV1/RV2 50k attenuator -> U7B/U7A TL074 follower -> 100k / 56k to +3V3 / 120k to GND. BAT54SW clamp, 1n -> U6B/U6A MCP6002 follower -> IO9/IO10.
Unpatched: RV = 0..+5 V knob. Patched: RV scales the CV, −5.9..+6.0 V -> 0..3.3 V.

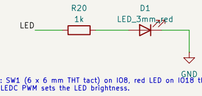


ES8388 CODEC (Bread Modular U4): I2S IO11–14/IO21 through 33R.
I2C IO47 (CCLK) / IO48 (CDATA), CE = GND (0x10). +3V3 -> R5 10R -> +3V3_8388.



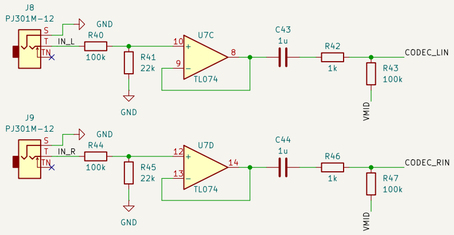
BUTTON



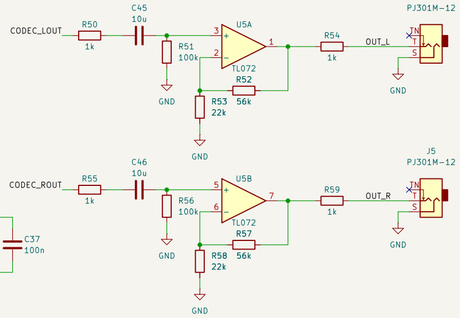


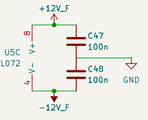
MODE: SW1 (6 x 6 mm THT tact) on IO8, red LED on IO18 through 1k (Bread Modular pins).
IO18 LEDC PWM sets the LED brightness.

AUDIO IN L/R: 100k / 22k (x0.18) -> U7C/U7D follower -> 1u -> 1k -> LIN1/RIN1 (100k to VMID).
+/−5 V -> +/−0.9 V; codec PGA sets the gain. No IN R normal: firmware handles mono.



AUDIO OUT L/R: LOUT1/ROUT1 -> 1k -> 10u -> U5A/U5B gain 3.55 (56k/22k) -> 1k 0.5 W -> J4/J5.
~1 Vrms codec full scale -> +/−5 V; the codec output volume trims the level.





Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S1 — Bread Modular 32bit–5v (ESP32-S3 + ES8388) for Eurorack 6HP: MIDI in, 2 audio i

Big Modular		
Sheet: /		
File: 32bit.kicad_sch		
Title: Big Modular — 32bit		
Size: A2	Date:	**Rev: S1**
KiCad E.D.A. 10.0.6		Id: 1/1